

Materials Bulk Upload Format

Use **Excel .xlsx** for bulk material upload.

Reason: materials are structured data in the app: course -> indexes -> slides -> content blocks -> files. Excel maps cleanly to database rows and can be validated before import. PPT is better for visual presentation, but extracting reliable indexes, slide titles, tables, lists, and attached files from PPT is inconsistent.

Sample Excel file: [materials-bulk-upload-sample.xlsx](./materials-bulk-upload-sample.xlsx)

Workbook Rules

The workbook must contain these sheets:

1. Indexes
2. Slides
3. Blocks
4. Assets optional

Do not rename sheets. Do not merge cells. Keep the first row as headers. Every row must have stable IDs so the importer can connect indexes, slides, blocks, and files.

The sample workbook locks the first header row and freezes it while scrolling. Team members should edit only data rows below the headers. Do not add, remove, or rename columns.

Sheet: Indexes

Column	Required	Rule
index_id	Yes	Unique key, for example IDX-001.
index_order	Yes	Number starting from 1.
index_title	Yes	Display name shown in Materials.
index_summary	No	Short internal summary.

Sheet: Slides

Column	Required	Rule
slide_id	Yes	Unique key, for example SLD-001.
index_id	Yes	Must match an Indexes.index_id.
slide_order	Yes	Number starting from 1 inside each index.
slide_title	Yes	Display title for the slide.
slide_summary	No	Optional slide summary.

Sheet: Blocks

Column	Required	Rule
block_id	Yes	Unique key, for example BLK-001.
slide_id	Yes	Must match a Slides.slide_id.
block_order	Yes	Number starting from 1 inside each slide.
block_type	Yes	One of heading, paragraph, assignment_note, bullet_list, nested_bullet_list, table.
block_title	No	Optional title shown above the block.
text	Conditional	Required for heading/paragraph/note/list blocks.
columns	Conditional	Required for table blocks. Separate columns with \ .
rows	Conditional	Required for table blocks. Separate cells with \ and rows with new lines.

List rules:

- bullet_list: put one bullet per line in text.
- nested_bullet_list: put one item per line in text; indent child items with two spaces.

Table example:

- columns: Term|Meaning
- rows: Anterior|Front surface then new line Posterior|Back surface

Sheet: Assets

Use this sheet only when slide files are already uploaded or stored in a known path.

```
| Column | Required | Rule |
| --- | --- | --- |
| asset_id | Yes | Unique key, for example AST-001. |
| slide_id | Yes | Must match a Slides.slide_id. |
| asset_order | Yes | Number starting from 1 inside each slide. |
| asset_type | Yes | One of image, video, document, link. |
| title | No | File name or display title. |
| url | Yes | Existing uploaded URL, external URL, or server-imported file path. |
```

Validation Rules

- IDs must be unique inside each sheet.
- All references must exist: Slides.index_id, Blocks.slide_id, and Assets.slide_id.
- Orders must be numeric and must not duplicate inside the same parent.
- Empty rows are ignored.
- Unsupported block_type or asset_type should fail import.
- The importer should save as draft first. Publishing must be a separate action after review.

Recommended Workflow

1. Team fills the Excel file using the required sheets.
2. Admin/instructor uploads the workbook.
3. System validates the workbook.
4. System creates/updates draft course content.
5. User reviews the draft in Materials.
6. User clicks Publish only after review.

Complete Sample Workbook

Use the following sample exactly as a reference when creating the Excel file. Each table below is one Excel sheet.

Sample Sheet: Indexes

```
| index_id | index_order | index_title | index_summary |
| --- | ---: | --- | --- |
| IDX-001 | 1 | Basic to Anatomy & Physiology | Introduction to anatomy terms, body positions, and cavities. |
| IDX-002 | 2 | Integumentary System | Skin, hair, nails, and related structures. |
| IDX-003 | 3 | Blood and Circulatory System | Blood, heart, vessels, and circulation overview. |
```

How to enter:

- index_id must be short and unique.
- index_order controls display order in the Indexes list.
- index_title is what students see.

Sample Sheet: Slides

```
| slide_id | index_id | slide_order | slide_title | slide_summary |
| --- | --- | ---: | --- | --- |
| SLD-001 | IDX-001 | 1 | Contents | Overview of topics in the first index. |
| SLD-002 | IDX-001 | 2 | Medical terminology is the language of medicine | Explains why terminology matters. |
| SLD-003 | IDX-001 | 3 | Body Views and Directional Terms | Directional body terms with meanings. |
```

```
| SLD-004 | IDX-001 | 4 | Body Cavities | Dorsal and ventral body cavities. |
| SLD-005 | IDX-002 | 1 | Functions of the Skin | Main functions of the integumentary system. |
```

How to enter:

- index_id must match a row from the Indexes sheet.
- slide_order starts again from 1 for each index.
- One slide can have many content blocks and many assets.

Sample Sheet: Blocks

```
| block_id | slide_id | block_order | block_type | block_title | text | columns | rows |
| --- | --- | ---: | --- | --- | --- | --- | --- |
| BLK-001 | SLD-001 | 1 | heading | | Contents | | |
| BLK-002 | SLD-001 | 2 | nested_bullet_list | | Medical Terminology:\n Prefix\n Suffix\n Root
Word\nWord Building\nBasic Human Anatomy and Physiology. | | |
| BLK-003 | SLD-002 | 1 | paragraph | Why terminology matters | Medical terminology is the standard language
used by healthcare professionals to describe the body, diagnosis, and treatment. | | |
| BLK-004 | SLD-002 | 2 | bullet_list | Key Points | Improves communication\nReduces confusion\nSupports
accurate documentation | | |
| BLK-005 | SLD-003 | 1 | table | Directional Terms | | Term\|Meaning | Anterior (Ventral)\|Front
surface\nPosterior (Dorsal)\|Back surface\nSuperior\n|Above\nInferior\n|Below |
| BLK-006 | SLD-004 | 1 | paragraph | Body cavities | Body cavities are spaces within the body that protect
internal organs, support organs, and separate organs. | | |
| BLK-007 | SLD-004 | 2 | nested_bullet_list | | Dorsal Body Cavity\n Cranial cavity: contains the
brain.\n Spinal cavity: contains the spinal cord.\nVentral Body Cavity\n Thoracic cavity: contains heart
and lungs.\n Abdominopelvic cavity: contains digestive and reproductive organs. | | |
| BLK-008 | SLD-005 | 1 | bullet_list | Functions | Protection\nTemperature regulation\nSensation\nVitamin D
synthesis | | |
```

Important entry rules:

- In Excel, line breaks inside a cell can be entered with Alt + Enter on Windows or Option + Command + Enter on Mac.
- For nested_bullet_list, child items must start with two spaces.
- For table, keep text empty and fill columns and rows.
- In columns, separate each column with |.
- In rows, separate each cell with | and each row with a line break.

Table block clean example:

```
| block_type | columns | rows | | |
|---|---|---|---|---|
| table | Term\|Meaning | Anterior\n|Front surface\nPosterior\n|Back surface |
```

Nested list clean example:

text

Dorsal Body Cavity

Cranial cavity: contains the brain.

Spinal cavity: contains the spinal cord.

Ventral Body Cavity

Thoracic cavity: contains heart and lungs.

Sample Sheet: Assets

```
| asset_id | slide_id | asset_order | asset_type | title | url |
| --- | --- | ---: | --- | --- | --- |
| AST-001 | SLD-003 | 1 | image | Body views.png | /uploads/content/course-id/body-views.png |
| AST-002 | SLD-004 | 1 | image | Dorsal body cavity.png | /uploads/content/course-id/dorsal-body-cavity.png
|
| AST-003 | SLD-004 | 2 | image | Ventral body cavity.png | /uploads/content/course-id/ventral-body-
cavity.png |
| AST-004 | SLD-005 | 1 | document | Skin notes.pdf | /uploads/content/course-id/skin-notes.pdf |
```

How to enter:

- Use image for .png, .jpg, .jpeg, .webp.
- Use document for .pdf, .doc, .docx, .ppt, .pptx.
- Use video for .mp4, .mov, .webm.

- Use link when the url is an external link.
- asset_order controls file carousel order inside the slide.

Minimal Example

If the team wants to test with the smallest valid workbook, use this:

Indexes

```
| index_id | index_order | index_title | index_summary |
| --- | ---: | --- | --- |
| IDX-001 | 1 | Anatomy Basics | Basic introduction. |
```

Slides

```
| slide_id | index_id | slide_order | slide_title | slide_summary |
| --- | --- | ---: | --- | --- |
| SLD-001 | IDX-001 | 1 | Introduction | First slide. |
```

Blocks

```
| block_id | slide_id | block_order | block_type | block_title | text | columns | rows |
| --- | --- | ---: | --- | --- | --- | --- | --- |
| BLK-001 | SLD-001 | 1 | heading | | Introduction | | |
| BLK-002 | SLD-001 | 2 | paragraph | | Welcome to Anatomy Basics. | | |
```

Assets can be empty or omitted for this minimal test.